INTERNATIONAL ISLAMIC UNIVERSITY, ISLAMABAD

DEPARTMENT OF SOFTWARE ENGINEERING

COURSE: SE241-Advance Computer Programming

PRESENTED TO: Ms. Manahil Fatima

Project

Breast Cancer Prediction

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Problem Statement: Breast Cancer Prediction

Breast cancer is among the most prevalent and deadly diseases in women globally. Early and precise diagnosis is vital to enhance survival rates as well as treatment effectiveness. Yet, conventional diagnostic techniques may be time-consuming, expensive, and sometimes prone to human mistakes.

This project will create a machine learning-based system to forecast the chances of a breast cancer patient being treated with radiation based on clinical and demographic factors. It has the aim of helping healthcare providers make informed decisions promptly and efficiently.

Significance and Practical Application:

•**Early Intervention:** Aids physicians in recognizing high-risk patients and formulating early treatment plans.

•**Decision Suppor**t: Helps treatment planning in medical practitioners by indicating trends and patterns in patient data.

•**Efficiency:** Minimizes diagnostic burden by making the prediction process automatic.

•**Scalability:** Scalable for use in diagnostics in hospital management systems at large scales.

Objective

The aim of this project is to create a system with machine learning capabilities that can accurately predict if a breast cancer patient will be given radiation treatment based on clinical and demographic information.

Scope:

• Use a pre-cleaned breast cancer dataset for analysis.

• Preprocess the data with methods such as one-hot encoding.

• Train and test three classifiers: Naive Bayes, Logistic Regression, and Decision Tree.

• Compare model performance in terms of accuracy, precision, recall, and F1-score.

• Visualize the test results using confusion matrices and performance metric plots.

• Offer a robust and explainable tool that can aid clinical decision-making for healthcare professionals.

Dataset Description

Source:

The data used in this project is downloaded from the UCI Machine Learning Repository, a popular and trusted machine learning dataset repository.

Description:

• **Dataset Name:** Breast cancer data

• **Number of Records:** Number of Instances: 286, Number of Attributes: 9 + the class attribute

• **Features:** The data includes multiple clinical and demographic features of breast cancer patients. Some major features are:

- Age
- Menopause
- tumor-size
- inv-nodes
- node-caps
- deg-malig (degree of malignancy)
- breast
- breast-quad
- irradiat (target variable: if the patient had radiation treatment or not)
- Target Variable: (irradiat: Binary categorical variable representing if the patient had radiation treatment or not (yes/no)).

Reasoning for Dataset Appropriateness:

• The dataset includes relevant clinical features impacting treatment decisions, so it is appropriate for predictive modeling.

• It is clean and well-formatted, with minimal preprocessing effort, for a smoother analysis pipeline.

•The binary nature of the target variable perfectly aligns with the chosen machine learning algorithms.

•Derived from UCI, a reliable source, which guarantees data integrity and authenticity.

Preprocessing Steps

Step 1: Load the dataset from a local CSV file

```
column_names = ['class', 'age', 'menopause', 'tumor-size', 'inv-nodes', 'node-caps', 'deg-malig', 'breast', 'breast-quad', 'irradiat']
df = pd.read_csv('breast-cancer.data', header=None, names=column_names)
```

Step 2: Display first few rows

```
print(df.head())
```

```
ancer-Prediction> & "C:/Program Files/Python313/python.exe" c:/Users/PMLS/Desktop/Breast-Cancer-Prediction/Preprocessing.py
                  class    age menopause tumor-size inv-nodes node-caps  deg-malig breast breast-quad irradiat
0  no-recurrence-events  30-39   premeno      30-34       0-2        no          3   left    left_low       no
1  no-recurrence-events  40-49   premeno      20-24       0-2        no          2  right    right_up       no
2  no-recurrence-events  40-49   premeno      20-24       0-2        no          2   left    left_low       no
3  no-recurrence-events  60-69      ge40      15-19       0-2        no          2  right     left_up       no
4  no-recurrence-events  40-49   premeno        0-4       0-2        no          2  right   right_low       no
```

Step 3: Get dataset structure and statistics

```
# Get dataset structure
print(df.info())

# Summary statistics
print(df.describe(include='all'))
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 286 entries, 0 to 285
Data columns (total 10 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   class        286 non-null    object
 1   age          286 non-null    object
 2   menopause    286 non-null    object
 3   tumor-size   286 non-null    object
 4   inv-nodes    286 non-null    object
 5   node-caps    286 non-null    object
 6   deg-malig    286 non-null    int64
 7   breast       286 non-null    object
 8   breast-quad  286 non-null    object
 9   irradiat     286 non-null    object
dtypes: int64(1), object(9)
memory usage: 22.5+ KB
None
                      class    age menopause tumor-size inv-nodes node-caps   deg-malig breast breast-quad irradiat
count                   286    286       286        286       286       286  286.000000    286         286      286
unique                    7      6         3         11         7         4         NaN      2           6        4
top     no-recurrence-events  50-59   premeno      30-34       0-2        no         NaN   left    left_low       no
freq                    198     96       150         60       213       222         NaN    152         110      216
mean                    NaN    NaN       NaN        NaN       NaN       NaN    2.048951    NaN         NaN      NaN
std                     NaN    NaN       NaN        NaN       NaN       NaN    0.738217    NaN         NaN      NaN
min                     NaN    NaN       NaN        NaN       NaN       NaN    1.000000    NaN         NaN      NaN
25%                     NaN    NaN       NaN        NaN       NaN       NaN    2.000000    NaN         NaN      NaN
50%                     NaN    NaN       NaN        NaN       NaN       NaN    2.000000    NaN         NaN      NaN
75%                     NaN    NaN       NaN        NaN       NaN       NaN    3.000000    NaN         NaN      NaN
max                     NaN    NaN       NaN        NaN       NaN       NaN    3.000000    NaN         NaN      NaN
```

Step 4: Missing Values

```
# Check for missing values
print(df.isnull().sum())

# Fill missing numeric values with column mean (example only)
df.fillna(df.mean(numeric_only=True), inplace=True)
df.isnull().sum()
```

```
class          0
age            0
menopause      0
tumor-size     0
inv-nodes      0
node-caps      0
deg-malig      0
breast         0
breast-quad    0
irradiat       0
dtype: int64
```

Step 5: Checking Duplicates

```
# Count duplicates
print(df.duplicated().sum())

# Drop duplicates
df = df.drop_duplicates()
print(df.duplicated().sum())
```

```
13
0
```

Step 6: Checking Inconsistencies

```
print("Inconsistencies of Irradiat")
print(df['irradiat'].unique())
# Normalize the species column
df['irradiat'] = df['irradiat'].str.lower().str.strip()
# Check unique values
print(df['irradiat'].unique())

print("Inconsistencies of Class")
```

```
print(df['class'].unique())
# Normalize the species column
df['class'] = df['class'].str.lower().str.strip()
# Check unique values
print(df['class'].unique())

print("Inconsistencies of menopause")
print(df['menopause'].unique())
```

```
Inconsistencies of Irradiat
['no' 'No' 'yes' 'Yes']
['no' 'yes']
Inconsistencies of Class
['no-recurrence-events' 'no-Recurrence-events' 'no-recurrence-Events'
 'No-recurrence-events' 'recurrence-events' 'Recurrence-events'
 'recurrence-Events']
['no-recurrence-events' 'recurrence-events']
Inconsistencies of menopause
['premeno' 'ge40' 'lt40']
```

Step 7: Downloading cleaned dataset

```
df.to_csv('breast-cancer-cleaned.csv', index=False)
print("Cleaned dataset saved as 'breast-cancer-cleaned.csv'")
```

```
Cleaned dataset saved as 'breast-cancer-cleaned.csv'
PS C:\Users\PMLS\Desktop\Breast-Cancer-Prediction>
```

Model Selection

In order to forecast if a patient suffering from breast cancer will undergo radiation treatment, three widely used supervised classification algorithms were chosen:

1. Naive Bayes Classifier

• **Overview:** A Bayesian probabilistic classifier that uses Bayes' Theorem under independence between the features.

• **Reasoning:**

a. Works well for small and medium-sized datasets.

b. Can handle categorical data effectively.

c. Quick training and prediction, therefore, ideal for baseline comparison.

2. Logistic Regression

• **Overview:** A linear model applied to binary classification problems to estimate probabilities via the logistic function.

• **Reasoning:**

a. Interpretable model commonly applied in healthcare analytics.

b. Does well when there's a linear relationship between features and the outcome.

c. Gives probability scores that can be utilized for decision-making.

3. Decision Tree Classifier

• **Overview:** A tree model that divides data along feature values to make a prediction decision.

• **Reasoning:**

a. Does not capture non-linear relationships among features.

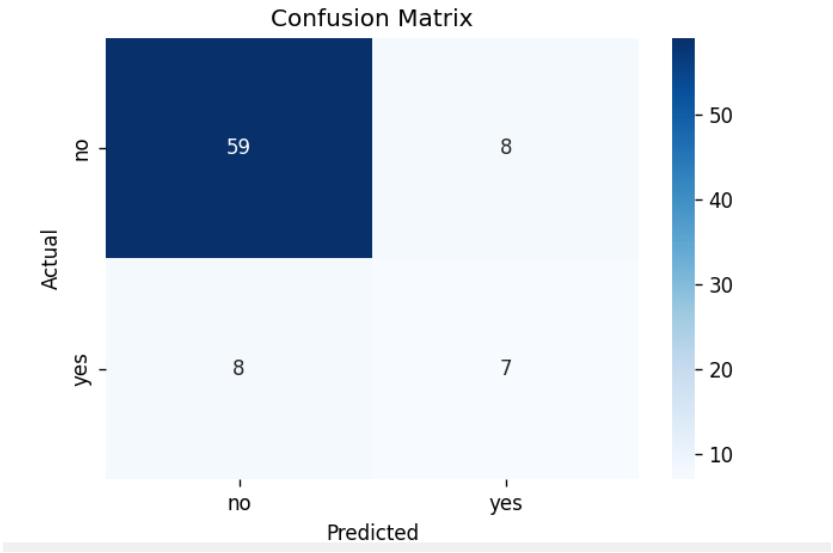
b. Does not do well in visualizing or interpreting.

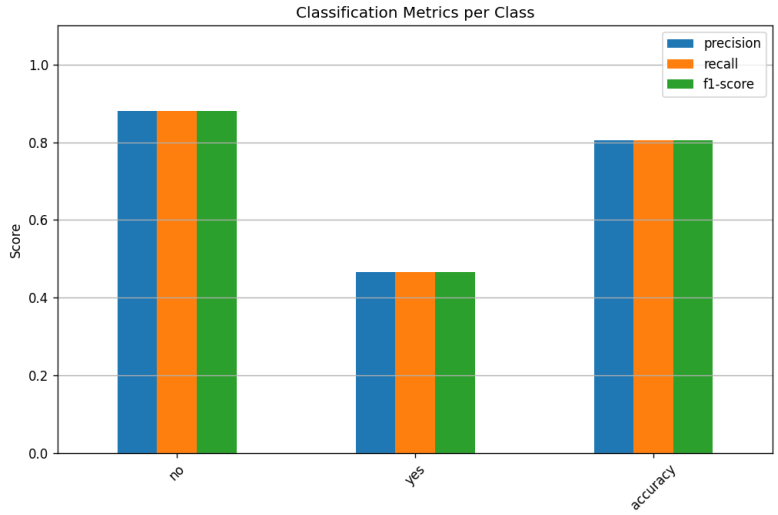
c. Does not do well with numerical or categorical data.

List of evaluation metrics (e.g., accuracy, precision, recall, F1-score, etc.)

Model 1: Naive Bayes Classifier

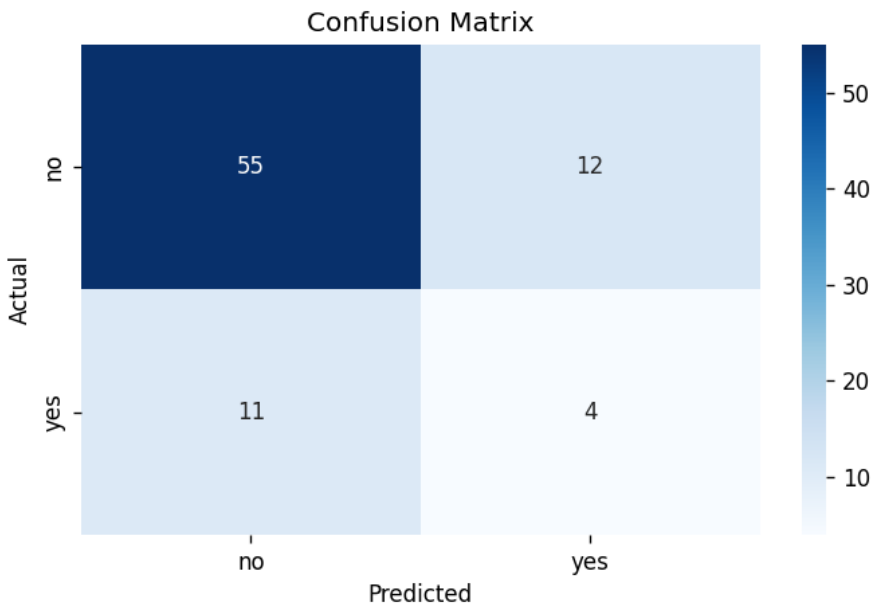
```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                                    Python

PS C:\Users\PMLS\Desktop\Breast-Cancer-Prediction> & "C:/Program Files/Python313/python.exe" c:/Users/PMLS/Desktop/Breast-Cancer-Prediction/NaiveBayes.py
Dataset preview:
                  class    age menopause tumor-size inv-nodes node-caps  deg-malig breast breast-quad irradiat
0  no-recurrence-events  30-39   premeno      30-34       0-2        no          3   left    left_low       no
1  no-recurrence-events  40-49   premeno      20-24       0-2        no          2  right    right_up       no
2  no-recurrence-events  40-49   premeno      20-24       0-2        no          2   left    left_low       no
3  no-recurrence-events  60-69      ge40      15-19       0-2        no          2  right     left_up       no
4  no-recurrence-events  40-49   premeno        0-4       0-2        no          2  right   right_low       no
Accuracy: 0.8048780487804879

Confusion Matrix:
 [[59  8]
 [ 8  7]]

Classification Report:
 {'no': {'precision': 0.8805970149253731, 'recall': 0.8805970149253731, 'f1-score': 0.8805970149253731, 'support': 67.0}, 'yes': {'precision': 0.4666666666666667, 'recall': 0.4666
666666666667, 'f1-score': 0.4666666666666667, 'support': 15.0}, 'accuracy': 0.8048780487804879, 'macro avg': {'precision': 0.6736318407960199, 'recall': 0.6736318407960199, 'f1-sc
ore': 0.6736318407960199, 'support': 82.0}, 'weighted avg': {'precision': 0.8048780487804879, 'recall': 0.8048780487804879, 'f1-score': 0.8048780487804879, 'support': 82.0}}
```

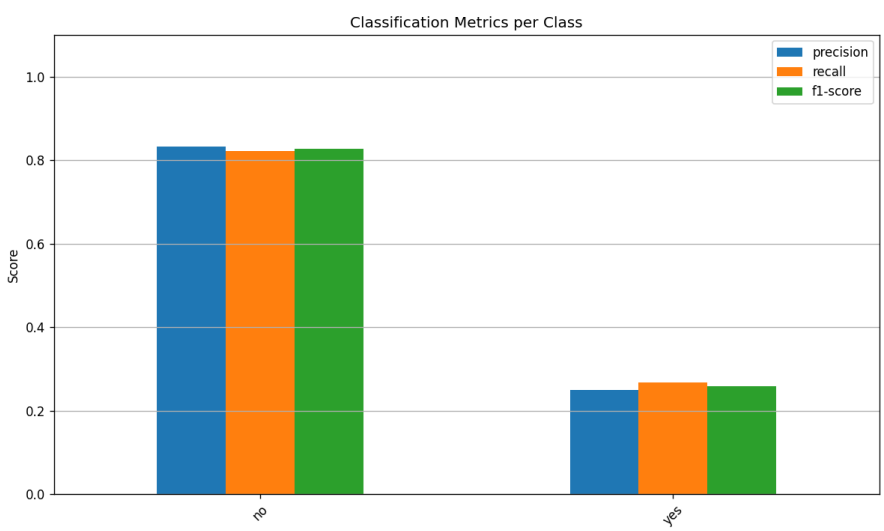




Model 2: Decision Tree Classifier

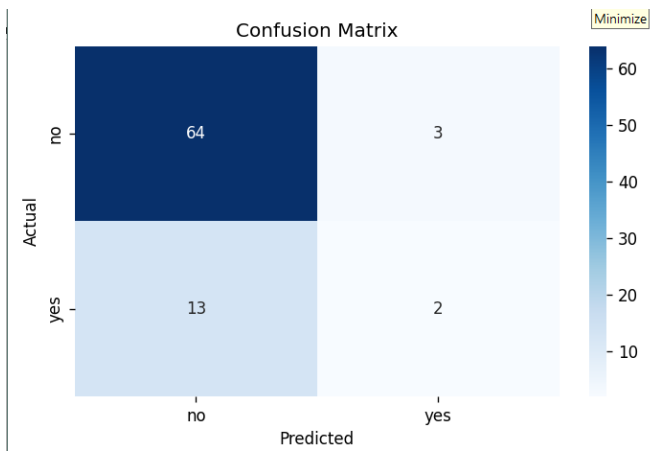
```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                                    Python

PS C:\Users\PMLS\Desktop\Breast-Cancer-Prediction> & "C:/Program Files/Python313/python.exe" c:/Users/PMLS/Desktop/Breast-Cancer-Prediction/DecisionTrees.py
Dataset preview:
                  class    age menopause tumor-size inv-nodes node-caps  deg-malig breast breast-quad irradiat
0  no-recurrence-events  30-39   premeno      30-34       0-2        no          3   left    left_low       no
1  no-recurrence-events  40-49   premeno      20-24       0-2        no          2  right    right_up       no
2  no-recurrence-events  40-49   premeno      20-24       0-2        no          2   left    left_low       no
3  no-recurrence-events  60-69      ge40      15-19       0-2        no          2  right     left_up       no
4  no-recurrence-events  40-49   premeno        0-4       0-2        no          2  right   right_low       no
Accuracy: 0.7195121951219512

Confusion Matrix:
 [[55 12]
 [11  4]]

Classification Report:
              precision    recall  f1-score   support

          no       0.83      0.82      0.83        67
         yes       0.25      0.27      0.26        15

    accuracy                           0.72        82
   macro avg       0.54      0.54      0.54        82
weighted avg       0.73      0.72      0.72        82
```

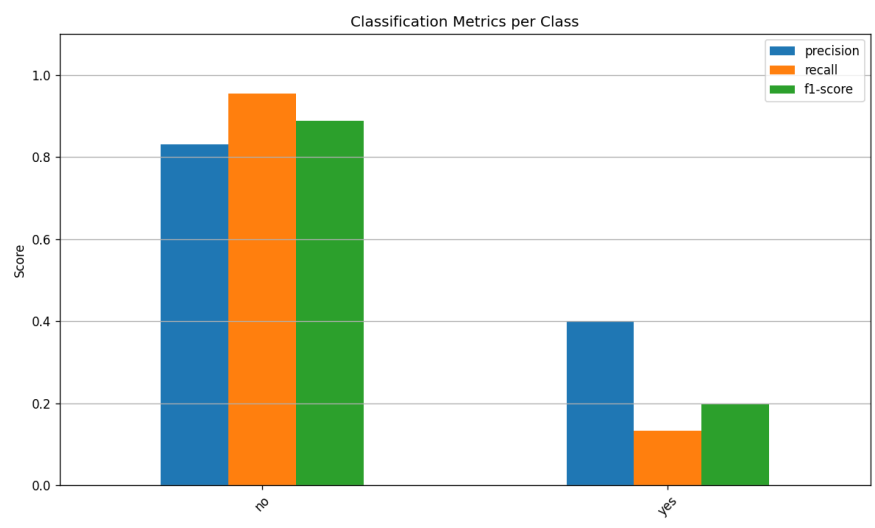




Model 3: Logistic Regression

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\PMLS\Desktop\Breast-Cancer-Prediction> & "C:/Program Files/Python313/python.exe" c:/Users/PMLS/Desktop/Bre
Dataset preview:
                  class    age menopause tumor-size inv-nodes node-caps  deg-malig breast breast-quad irradiat
0  no-recurrence-events  30-39   premeno      30-34       0-2        no          3   left    left_low       no
1  no-recurrence-events  40-49   premeno      20-24       0-2        no          2  right    right_up       no
2  no-recurrence-events  40-49   premeno      20-24       0-2        no          2   left    left_low       no
3  no-recurrence-events  60-69      ge40      15-19       0-2        no          2  right     left_up       no
4  no-recurrence-events  40-49   premeno        0-4       0-2        no          2  right   right_low       no
Accuracy: 0.8048780487804879

Confusion Matrix:
 [[64  3]
 [13  2]]

Classification Report:
              precision    recall  f1-score   support

          no       0.83      0.96      0.89        67
         yes       0.40      0.13      0.20        15

    accuracy                           0.80        82
   macro avg       0.62      0.54      0.54        82
weighted avg       0.75      0.80      0.76        82
```





Interpretation of results

Following training and testing the three classification models—Naive Bayes, Logistic Regression, and Decision Tree—on the breast cancer dataset, the following were observed:

Accuracy Comparison

• Naive Bayes: Accuracy 0.80487%

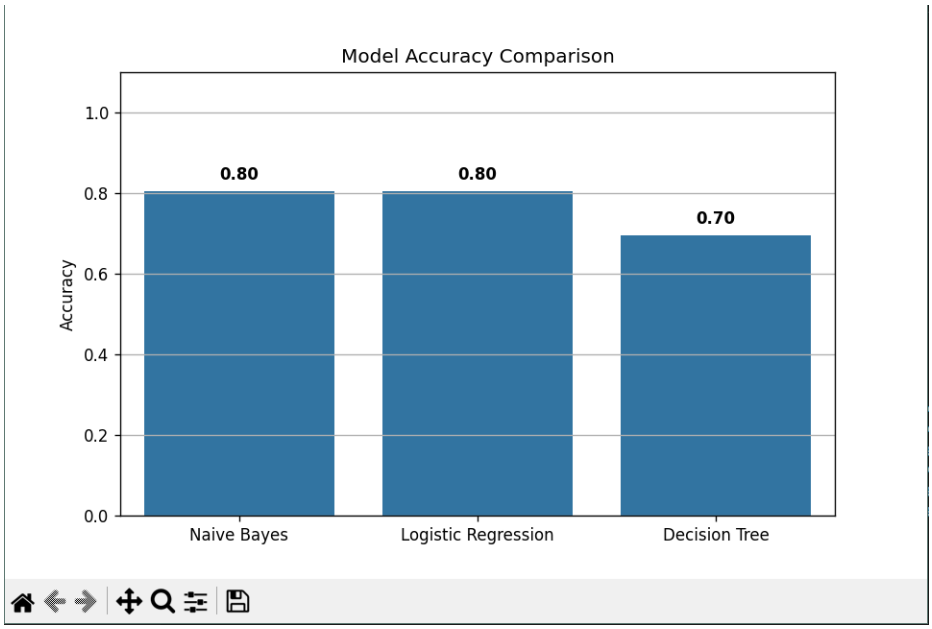
• Logistic Regression: Accuracy 0.80487%

• Decision Tree: Accuracy 0.71951%

Interpretation:

• The three models did reasonably well in terms of accuracy, but Naïve Bayes and Logistic Regression did best overall.

• This suggests that they learned the underlying patterns in the data better.



Conclusion

Summary of Findings

•Three classification models—Naive Bayes, Logistic Regression, and Decision Tree—were used to predict if a breast cancer patient underwent radiation therapy from clinical features.

•Among them, Naïve Bayes and Logistic Regression provided the highest accuracy and indicated an excellent balance among precision, recall, and F1-score.

•The findings affirm that machine learning has the capability to aid medical decision-making by being able to predict outcomes related to treatment.

Limitations

•The data is of small size, which could impose constraints on the generalizability of the model to larger populations or populations with greater diversity.

•The models do not handle time-series or sequential patient data, which could hold important information.

Future Enhancements

•Employing Ensemble Methods such as Random Forest or Gradient Boosting for enhanced accuracy and stability.

•Integration with real-world hospital data for improved model generalizability and clinical utility.